

MedTrack_DV

Dashboard Storyboard | Milestone 2 | Hospital Operations & Patient Analytics

This storyboard defines each of the four dashboards before Tableau development, following the Dashboard → Purpose → KPIs → Charts → Filters → Interactions structure.

1. Hospital Overview

"How is the hospital performing overall?" — primary source: *department_analytics_dataset* (trends), *hospital_overview_dataset* (detail)

PURPOSE

Give hospital management a single-page summary of overall performance: volume, occupancy, length of stay, readmissions, and bed usage, with drill-down into admission type, disease mix, and blood group demand.

KPIs

- Total Admissions
- Occupancy Rate
- Average Length of Stay
- Readmission Rate
- Bed Utilization Rate

CHARTS

Worksheet	Chart Type
KPI Strip (5 tiles)	Text / Big Number
Occupancy Trend	Area Chart
Admission vs Discharge	Dual-Line Chart
Bed Capacity by Department	Grouped Bar
Admissions by Type	Pie Chart
Admission by Department	Crosstab (% of total)
Disease Mix	Crosstab (% of total)
Blood Group Mix	Packed Bubble

FILTERS

- Department Selector (parameter — 6 clinical departments)
- Year Selector (parameter — 2020–2025 or All)

INTERACTIONS

- Both parameters drive every chart on the page simultaneously
- Navigation bar (Home / Patient Flow / Department Analytics / Resource Utilization)

2. Patient Flow

"How are patients moving through the hospital?" — primary source: patient_flow_dataset

PURPOSE

Show the full rhythm of hospital activity — admissions, discharges, and diagnostic-test traffic — together, including genuine cross-department movement into Radiology/Pathology that admission data alone cannot show.

KPIs

- (No KPI strip — this dashboard is chart-led; Dashboard 1 already carries the headline numbers)

CHARTS

Worksheet	Chart Type
Patient Volume by Event Type	Stacked Area
Peak Load by Day of Week	Heatmap
Home to Diagnostic Flow	Heatmap / Crosstab
Length of Stay Distribution	Histogram
Diagnostic Test Outcomes by Category	Crosstab
Diagnostic Coverage	Pie Chart

FILTERS

- Department Selector (parameter, shared with Dashboard 1)
- Year Selector (parameter, shared with Dashboard 1)
- Event Type (legend filter: Admission / Discharge / Diagnostic Test)

INTERACTIONS

- Same two global parameters keep this in sync with Dashboard 1
- Navigation bar to all other dashboards

3. Department Analytics

"Which departments are performing well and which need attention?" — primary source: `department_analytics_dataset`

PURPOSE

Compare all clinical departments side by side on the same set of KPIs, and surface the ICU efficiency finding (highest occupancy, lowest efficiency score, driven by longer average stay).

KPIs

- Admissions, Occupancy Rate, Avg LOS, Readmission Rate, and Department Efficiency Score — all computed per department

CHARTS

Worksheet	Chart Type
Department Scorecard	Highlight Table
Efficiency Score by Department	Sorted Bar
LOS Distribution by Department	Box-and-Whisker Plot
Readmission Rate by Department	Bar + Average Reference Line
Occupancy Trend by Department	Multi-Line Chart
Diagnostic Test Load (Radiology/Pathology)	Multi-Line Chart
Revenue by Department	Treemap

FILTERS

- Department Selector (parameter, set as Context Filter on Scorecard and Efficiency Score)
- Year Selector (parameter, set as Context Filter on the same two)

INTERACTIONS

- Context Filters ensure the Efficiency Score LOD calculations respect the Year selection
- Diagnostic Test Load intentionally excluded from Department Selector — Radiology/Pathology aren't in that parameter's list
- Navigation bar to all other dashboards

4. Resource Utilization

"How efficiently are hospital resources being used?" — primary source: resource_utilization_dataset

PURPOSE

Cover the three resource types the hospital manages — beds, staff, and drug inventory — clearly separated by their true grain: beds are a real daily trend; staff and drugs are current snapshots.

KPIs

- Bed Utilization Rate (daily trend)
- Capacity vs Demand (Max beds vs Average use)
- Shortage rate and severity across current drug stock

CHARTS

Worksheet	Chart Type
Bed Utilization Trend	Multi-Line Chart
Capacity vs Demand	Grouped Bar (Max / Average)
Staff Allocation by Department & Shift	Grouped Bar
Shortage Rate by Drug Category	Sorted Bar
Drug Shortage Watch	Sorted Bar, colored by severity

FILTERS

- Resource Type (per-chart: Bed / Staff / Drug Inventory)
- Department Selector — applies to Bed and Staff charts only
- Year Selector — applies to Bed charts only (Staff and Drug Inventory have no date axis)

INTERACTIONS

- Staff and Drug charts are explicitly labeled as current snapshots, not trends
- Navigation bar to all other dashboards